FMCW GPR Signal-Processing and Data-Acquisition Design

805–868 MHz FMCW GPR | Mixer → IF → ADC (AD9226) → Raspberry Pi 4 SMI/DMA → FFT → Radargram

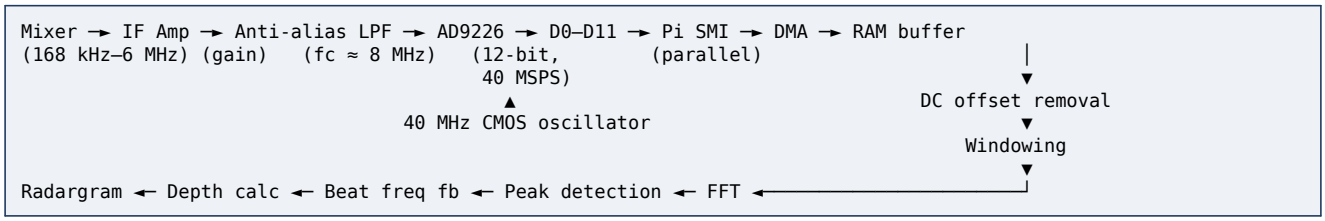
Parameter	Target
RF sweep / centre / bandwidth	805–868 MHz / 836.5 MHz / 63 MHz
Sweep	1 MHz triangular
IF / beat range	≈168 kHz – 6 MHz
ADC	AD9226, 12-bit, 40 MSPS, 40 MHz clock, 12-bit parallel
Processor / capture	Raspberry Pi 4B, SMI + DMA, FFT/DSP on the Pi

Key design risks (read first)

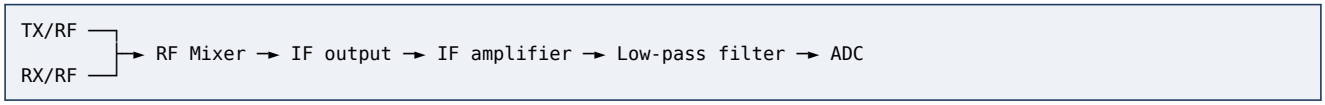
Verification status: Part specifications, prices and the GPIO mapping in this document were written from memory without live datasheet or distributor lookups. Items marked **verify** must be checked against manufacturer datasheets, the BCM2711 documentation and current Indian distributor listings before ordering or wiring.

- **Resolution is set by bandwidth, not FFT size.** Range resolution = c/(2B√εr): 2.38 m in air and about 0.79 m at εr = 9. A 0.5 µs ramp gives only 2 MHz beat resolution (the same limit in frequency). Large FFTs interpolate; they do not add resolution.
- **Each ramp holds only 20 samples** (0.5 µs at 40 MSPS). A 168 kHz beat completes about 0.08 cycle per ramp and 6 MHz completes 3. Derive the sweep trigger from the same 40 MHz clock divided by 40 so every sweep is exactly 40 samples, average many sweeps coherently, then zero-pad and FFT.
- **Pi 4 SMI at 40 MSPS is unverified and is the largest risk.** The raw stream is 80 MB/s with samples stored in 16-bit words. The SMI read strobe is not synchronous with a free-running ADC clock, so samples can be duplicated or skipped. Verify against BCM2711 SMI timing and published SMI-ADC projects. Fallbacks: a lower sample rate (20 MSPS still covers 6 MHz) or a FIFO/CPLD buffer in front of the Pi.

1 and 9. Complete signal-processing block diagram



2. Mixer output section



Expected IF output. The mixer multiplies TX and RX. The desired term is the difference frequency fb = S·τ, in the 168 kHz–6 MHz range. The sum term (about 1.67 GHz) and any RF/LO leakage are far outside this band and must be heavily attenuated.

Why about 830 MHz at the IF port indicates a problem. It means RF or LO is leaking through: poor port isolation, swapped or wrong mixer ports, an unterminated or mismatched port, or a measurement-setup artifact (probe or long ground lead acting as an antenna). It also matters for the ADC: 836.5 MHz folds to 3.5 MHz at 40 MSPS (836.5 − 800 = 36.5; 40 − 36.5 = 3.5), and 830 MHz folds to 10 MHz. Residual RF leakage can therefore appear as a false in-band beat unless the filter and shielding

remove it.

Measurement procedure

- Terminate the mixer RF and LO ports in 50 Ω at the specified drive levels. Look at the IF port with a spectrum analyzer from 100 kHz to 1.5 GHz, using a DC block and attenuator for protection.
- Record any energy near 805–868 MHz relative to the IF tones. It should be tens of dB below.
- Use an oscilloscope with a 50 Ω feedthrough termination and a short ground spring. Long ground leads create false RF pickup.
- Repeat with the RX port terminated, then the TX port terminated, to locate the leakage path.
- Confirm the port pin-out against the mixer datasheet.

3. IF amplifier (168 kHz–6 MHz)

Choose bandwidth at least 10× the needed band at your operating gain. Specs are from memory (**verify**). Prices are unverified estimates.

Part	Bandwidth	Gain	Supply	Noise	Price (INR)	Status
LMH6624	≈1.5 GHz GBW (min gain 10)	10–20×	±2.5–6 V	≈0.9 nV/√Hz	₹300–1,500	Recommended
ADA4898-1	≈65 MHz unity gain	Flexible	±5–16 V	≈0.9 nV/√Hz	₹300–1,500	Best performance
OPA656	≈230 MHz GBW	Flexible	±5 V	Low (FET input)	₹300–1,500	Optional
AD8065	≈145 MHz GBW	Flexible	5–24 V	Moderate	₹300–1,500	Budget

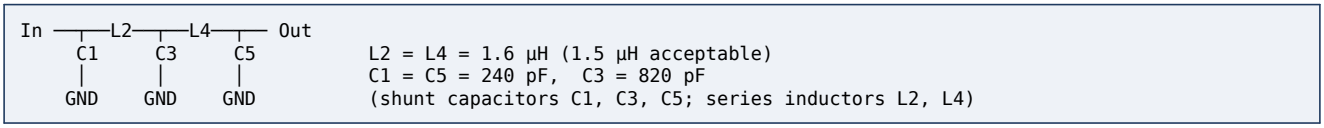
- Match the input to the mixer IF port impedance (check its datasheet; many IF ports are 50 Ω class).
- AC-couple with a corner around 20–50 kHz to remove DC offset and self-mixing products.
- Slew rate must be at least ≈40 V/µs for 2 Vpp at 6 MHz (2π·6 MHz·1 V ≈ 38 V/µs).

```
Mixer ─► IF amplifier ─► LPF ─► ADC
```

4. Anti-aliasing filter

The filter must pass 168 kHz–6 MHz. Anything above fs − 6 MHz = 34 MHz folds into the useful band, so the filter's main job is strong attenuation above 34 MHz; the transition from 6 to 34 MHz can be gentle. Content between 6 and 34 MHz folds only into the 6–20 MHz region that is ignored digitally.

Recommended: 5th-order Butterworth LC low-pass, fc = 8 MHz, 50 Ω (standard prototype values 0.618, 1.618, 2.000, 1.618, 0.618).



Item	Value
Filter type / order	Butterworth, 5th order, LC ladder, 50 Ω
Cutoff	8 MHz (−3 dB)
Passband	168 kHz – 6 MHz (≈ −0.24 dB at 6 MHz)
Stopband	Above 34 MHz (alias band into 0–6 MHz)
Attenuation	≈40 dB at 20 MHz; ≈63 dB at 34 MHz (ideal); 830 MHz limited by parasitics and coupling
Components	C0G/NP0 capacitors; inductors with SRF well above 100 MHz
Active alternative	Sallen-Key sections: GBW ≥ 100–200 MHz, slew ≥ 100 V/µs, noise < 3 nV/√Hz. LC is usually better at 8 MHz.

Add an ADC driver (AD8138 or ADA4930) with a series resistor and a small capacitor at the ADC input. Shielding and layout matter as much as the filter for rejecting 830 MHz leakage.

5. AD9226 ADC

Typical values from memory (**verify** against the Analog Devices datasheet).

Parameter	Value
Resolution	12 bit
Maximum sampling rate	65 MSPS
Project sampling rate	40 MSPS
Analog input	≈2 Vpp span (internal reference); single-ended or differential
Supply	5 V analog; digital 3–5 V (use 3.3 V DRVDD)
Digital output	3.3 V CMOS with DRVDD = 3.3 V
Channels	1
Output format	Parallel D0–D11
Clock input	Single-ended CMOS, ≈50% duty cycle
SNR	≈69 dB (typical)
ENOB	≈11 bits (typical)
Input bandwidth	≈750 MHz full-power (typical)
Power	≈475 mW (typical)
Package / modules	28-SSOP; many low-cost breakout modules exist

- **Why 40 MSPS suits 6 MHz:** 40 MHz / 6 MHz = 6.67 samples per IF cycle at the top of the band; about 238 samples per cycle at 168 kHz. Nyquist is 20 MHz, leaving ample margin.
- **Raw data rate:** 12 bits × 40 MSPS = 480 Mbit/s = 60 MB/s. Stored in 16-bit words: 80 MB/s.

```
IF signal → AD9226 → D0–D11 → Raspberry Pi
```

6. 40 MHz ADC clock

For a 6 MHz input at ≈69 dB SNR the allowed rms jitter is about $1/(2\pi \cdot 6\ \text{MHz} \cdot 10^{69/20}) \approx 9$ ps. Compare candidates against that figure. Specs and prices are estimates (**verify**).

Component	Frequency	Supply	Output	Stability	Jitter	Price (INR)	Recommendation
Crystal + own oscillator	40 MHz	—	Needs circuit	±20–50 ppm	Circuit-dependent	Low	Not preferred
CMOS oscillator (Epson / SiTime / Abracon families)	40.000 MHz	3.3 V	CMOS	±20–50 ppm	≈1 ps rms typical	₹200–600	Preferred
TCXO (e.g. SiT5356 or equivalent; confirm 40 MHz option)	40 MHz	3.3 V	CMOS / clipped sine	≈±0.5–2.5 ppm	Low	₹1,000–3,000	Best performance
Si5351	Programmable	3.3 V	CMOS	Crystal-dependent	Tens of ps p-p (marginal)	₹200–500	Not for ADC clock

```
40 MHz oscillator → (22–33 Ω series R) → AD9226 CLK
```

Why DS1307/DS3231 are not suitable: they are real-time-clock chips built around a 32.768 kHz crystal that keep time over I²C. They cannot generate 40 MHz; the DS3231 square-wave output tops out at 32.768 kHz.

7. Raspberry Pi SMI + DMA

```
AD9226 D0–D11 → Pi GPIO/SMI → SMI hardware (FIFO) → DMA → RAM buffer
```

- **SMI** (Secondary Memory Interface) is a Broadcom peripheral providing an 8/16-bit parallel bus with address lines and read/write strobes with programmable timing.
- **DMA** (Direct Memory Access) is a hardware engine that moves data from the SMI FIFO to RAM without CPU involvement.
- **Why Python GPIO polling fails:** polling pins in Python manages tens of kHz to a few hundred kHz with irregular OS-scheduled timing, hundreds of times too slow.

- **Capture flow:** SMI issues read cycles → latches the parallel pins into the FIFO → DMA writes a continuous chain of blocks to RAM.
- **Post-processing:** a C/C++ program (or Python/NumPy over the buffer) unpacks the 12 bits, removes DC, windows and FFTs.
- **GPIO mapping (verify before wiring):** from recollection of the BCM2835/BCM2711 peripheral tables, SMI is on ALT1 with SA5–SA0 on GPIO0–5, SOE/SWE on GPIO6–7 and SD0–SD15 on GPIO8–23. This is not confirmed; check the official BCM2711 ARM Peripherals document and Raspberry Pi pinout.
- **3.3 V limitation:** Pi GPIO is 3.3 V only and not 5 V tolerant. Run AD9226 DRVDD at 3.3 V, add 330 Ω series resistors on data lines, and level-shift any 5 V signal.

8 and 14. BOM and shopping list

All prices are unverified INR estimates and need verification with current distributor listings. Part numbers marked as families need the exact suffix confirmed.

Component	Recommended part	Specification	Qty	Purpose	Approx. price	Priority
ADC	AD9226ARSZ (best) or generic AD9226 module (budget)	12-bit, 65 MSPS	1	Digitize IF	₹1,500–4,000	Essential
Clock	40 MHz CMOS oscillator (budget/recommended); TCXO (best)	3.3 V CMOS	1	ADC clock	₹200–3,000	Essential
IF amp	LMH6624 (recommended); ADA4898-1 (best); AD8065 (budget)	See section 3	1–2	Gain	₹300–1,500	Essential
LPF parts	C0G caps 240 pF ×2, 820 pF ×1; inductors 1.6 µH ×2	8 MHz Butterworth	1 set	Anti-alias	₹100–300	Essential
ADC driver	AD8138 or ADA4930	Differential	1	Drive ADC	₹500–1,500	Recommended
Series resistors	330 Ω	0603, 1%	14	GPIO protection	₹20	Essential
SMA connectors	Edge-mount SMA	50 Ω	3–4	RF/IF I/O	₹150–400	Essential
Coax	RG316 or RG174 with SMA	50 Ω	2–3	Interconnect	₹300–800	Essential
Raspberry Pi	Raspberry Pi 4B 2GB	BCM2711	1	SMI + DSP	₹4,500–6,500	Essential
MicroSD	32 GB A1/A2	—	1	OS	₹400–800	Essential
Pi power supply	Official 5.1 V 3 A USB-C	—	1	Power	₹600–1,000	Essential
Decoupling	100 nF + 10 µF per supply pin	Ceramic	≈20	Noise	₹100	Essential
Regulators	Low-noise LDO (e.g. LT1763 / TPS7A family)	5 V, 3.3 V	2	Clean analog supply	₹200–800	Recommended
PCB	Custom 4-layer, solid ground plane	—	1	Layout	₹1,500–4,000	Recommended
Shielding	Shield can or copper tape	—	1	Isolate IF/ADC from RF	₹200–600	Recommended

10. Digital signal processing

```
frame = ADC_words & 0x0FFF            # unpack 12 bits
frame = average(frames over N sweeps) # coherent, sweep-synchronous
frame -= mean(frame)                  # DC removal
(optional) bandpass 168 kHz – 6 MHz
frame *= window(Hann)
spec  = |FFT(frame, zero-padded to N)|
spec[bins < 168 kHz] = 0              # reject DC / near-DC leakage
mask known leakage lines              # e.g. aliased RF leakage bin
peaks = find_peaks(spec, threshold)
fb    = bin_index * fs / N
d     = fb * c / (2 * S * sqrt(eps_r))
```

FFT size N	Bin spacing (40 MSPS)	Record length	Depth per bin, air	Depth per bin, εr = 9
1024	39.06 kHz	25.6 µs	4.6 cm	1.5 cm
4096	9.77 kHz	102.4 µs	1.2 cm	0.39 cm
16384	2.44 kHz	409.6 µs	0.29 cm	0.10 cm
65536	610 Hz	1.64 ms	0.07 cm	0.02 cm

Recommendation: 4096 points (or 4000 for exactly 100 sweeps) on sweep-synchronous averaged data. It is cheap on the Pi and gives fine interpolation. Larger sizes add compute but no real resolution, because the physical limit is the 2 MHz ramp resolution.

11. FMCW beat-frequency equation

- $\tau = 2d/v$, $v = c/\sqrt{\varepsilon r}$, $fb = 2 \cdot S \cdot d \cdot \sqrt{\varepsilon r} / c$, therefore $d = fb \cdot c / (2 \cdot S \cdot \sqrt{\varepsilon r})$.
- Sweep slope $S = 63\ \text{MHz} / 0.5\ \mu\text{s} = 1.26 \times 10^{14}$ Hz/s (one ramp of the 1 µs triangle).
- $fb \approx 840$ kHz per metre in air, scaling with $\sqrt{\varepsilon r}$.
- The real slope must come from the measured VCO tuning curve. A real VCO is not perfectly linear, so calibrate against it or linearize the ramp.

Depth	εr = 1	εr = 4	εr = 9	εr = 16
0.1 m	84 kHz	168 kHz	252 kHz	336 kHz
0.25 m	210 kHz	420 kHz	630 kHz	840 kHz
0.5 m	420 kHz	840 kHz	1.26 MHz	1.68 MHz
1 m	840 kHz	1.68 MHz	2.52 MHz	3.36 MHz
2 m	1.68 MHz	3.36 MHz	5.04 MHz	6.72 MHz

This matches the 168 kHz lower bound (0.1 m at εr = 4, or 0.2 m in air). The 6 MHz upper bound corresponds to about 2.4 m at εr = 9.

12. Radargram processing

```
Repeated GPR scans → ADC acquisition → FFT per scan → Beat-frequency spectrum
 → Depth conversion → Background/clutter removal → Amplitude normalization
 → 2D matrix → Radargram
```

- **X-axis:** scan index or position.
- **Y-axis:** depth.
- **Pixel intensity:** reflected signal amplitude, after subtracting the mean (or a moving average) across scans to remove static clutter and direct leakage.

13. Final signal-processing architecture

```
ANALOG
  Mixer → IF amp → LPF
          LMH6624     5th-order Butterworth
          10–20×,     fc 8 MHz, 50 Ω
          >60 MHz BW
DATA ACQUISITION
  LPF → AD9226 ──────→ D0–D11 + 330 Ω → Pi SMI → DMA → RAM
        12-bit,   ▲                     3.3 V      80 MB/s   ≥128 KB
        40 MSPS   │                     GPIO
        DRVDD 3.3 V
                  └── 40 MHz CMOS osc (3.3 V, <1 ps rms jitter)
DSP
  RAM → DC removal → Hann window → FFT (4096) → Peak detect (>168 kHz)
      → fb → depth (S = 1.26e14 Hz/s, εr) → Radargram
```